FUNDAMENTALS OF C

C character set: Every programming language is a particular character set any by using this character set only we can develop the programs [software]. In C language we are using ASCII character set designed by IBM Corporation with 256 characters. In this we are having 52 alphabets [a-z, A-Z], 10 digits [0-9], 44 operators [+,-,*,…], 14 separators [, . ; : " " ' ', ( ) , { },…] and remain are special characters.

English language	C Language
26 alphabets	256 ASCII characters
English words	Keywords - 32
English sentences	Instructions
English paragraphs	Programs
English documents	Software

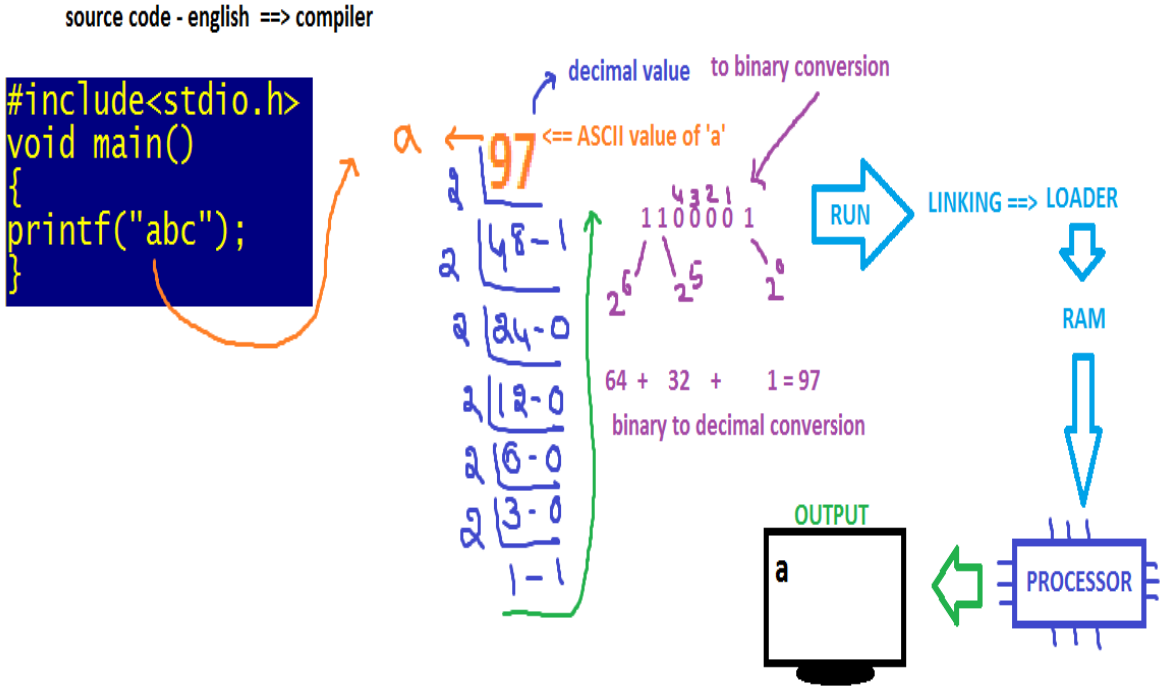
ASCII - American Standard Code for Information Interchange designed by IBM Corporation [International Business Machines].

Java – UNICODE character set – Universal Code – 65536 characters

We are using 4 type of number systems.

1. **Binary system - %d ← conversion characters / format specifiers**
2. **Decimal system - %d**
3. **Octal system - %o**
4. **Hexadecimal - %x**

Program compilation and execution process:



Some of the important ASCII values:

Characters	Ascii values
a-z	97-122
A-Z	65-90
0-9	48-57
*	42
+	43
Space	32
Esc	27
Tab space	9
Back space	8

C - TOKENS

The smallest individual words we are using in developing a c program are called c-tokens. They are of different types.

1. **Keywords: The system predefined / reserved words are called keywords. Each keyword is having certain meaning and we can't change this meaning. C comes with 32 words.**
   Eg: auto, break, const, continue, char, case, do, default, double, else, enum, extern, for, float, goto, int, long , while, switch, union,….